Extensibility of Poly-Catenane Structures

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1 INTRODUCTION

Poly-Catenanes are polymers made from a chain of mechanically interlocked rings. Due to the ring geometry of these polymer structures, we expect the poly-catenane chains to exhibit interesting and useful thermodynamic properties. Large chain poly-catenanes have yet to be developed in a laboratory setting. Nonetheless, we can use computer simulations of these polymers to study these theoretical thermodynamic properties. In this paper, we will find the extensibility of our poly-catenane structure using a Metropolis Monte Carlo simulation.

2 SIMULATION METHODS

The Metropolis Monte Carlo method is a way to simulate thermal fluctuations by either accepting or rejecting random moves based on how a system's energy changes. The following outlines the rough structure of how the Metropolis Monte Carlo method works when simulating the thermodynamics of the poly-catenane structure.

1. Initialize the State of the poly catenane system.
2. Propose a trial move
3. compute resulting ΔE
4. Accept if $\Delta E \leq 0$, or $\Delta E \geq 0$ with the probability $e^{-\beta \Delta E}$.
5. Update system variables and repeat.

Defining an initial state of the system entails defining all system and thermodynamic variables. The following are of interest:

- β: the inverse temperature coefficient, where $\beta = \frac{1}{KT}$.
- N: the number of catenane rings.
- L: The starting end-to-end distance of the center of our poly-catenane rings with the constraint $L < 2N - 2$.

The simulation method works since we are accepting and rejecting moves that correspond to physical thermodynamic laws derived through statistical mechanics. Since our system is held at a constant temperature, volume, and number of rings, we know that it can be represented as a Canonical Ensemble. We know a system will always want to be in the lowest possible energy state, so we always accept moves that lower the system's energy. If the system's energy increases as a result of a move, we have to accept it with the probability $e^{-B\Delta E}$, which is the Boltzmann Factor of our ensemble. This way, our poly catenane system always tends towards equilibrium.

In order to avoid our poly catenane rings becoming disconnected, regarding moves when the chain is at larger lengths, we add the following potential energy correction to our system:

$$U(d_{ij}) = \varepsilon e^{\frac{-d_{ij}}{\lambda}}$$

Where d_{ij} is the distance of rings i and j to each other.

The program also rejects any move in which rings become disconnected.

3 Method For Finding Extensibility

The extensibility of our poly-catenane structure is defined as:

$$(\frac{\partial L}{\partial F_x})_N$$

Or how the length of our poly-catenane changes with respect to the force being applied (in the x-direction), at a constant number of catenane rings. In order to calculate this, we will need to average the force exerted on the last polymer ring in our system once our system reaches equilibrium.

Even though we are finding out how the system's length changes as a function of the force applied, it is more intuitive to think about how the force changes with respect to a given length. This approach is more beneficial because we can actually define a set length L in our program, whereas defining a specific F would involve more work. Our goal is to find a Force vs.

Length Graph, in which we can curve fit in order to approximate our extensibility.

We have set up the program to loop through our Metropolis Monte Carlo simulation for lengths $3 \leq L \leq 17$ with the number of rings $N = 10$. These lengths will allow us to avoid some of the more extreme and most likely improbable properties for the extreme values on our length interval. Each of these runs will output the forces on the catenane chain in the x, y, and z directions, and the Energy for a given timestep. Data from each run is also exported in a comma-separated-value list (.csv) in order to do the data analysis in Python rather than C++, in which our simulation is coded. In Python, we

4 Results and Data Analysis

In order to know which time interval to average over, we will first look at the force vs time graph in the x-direction. Once the graph settles around a particular force, we know that the system has reached equilibrium.

Through inspection, we can estimate that most lengths will be in equilibrium after $t = 20,000$ steps.

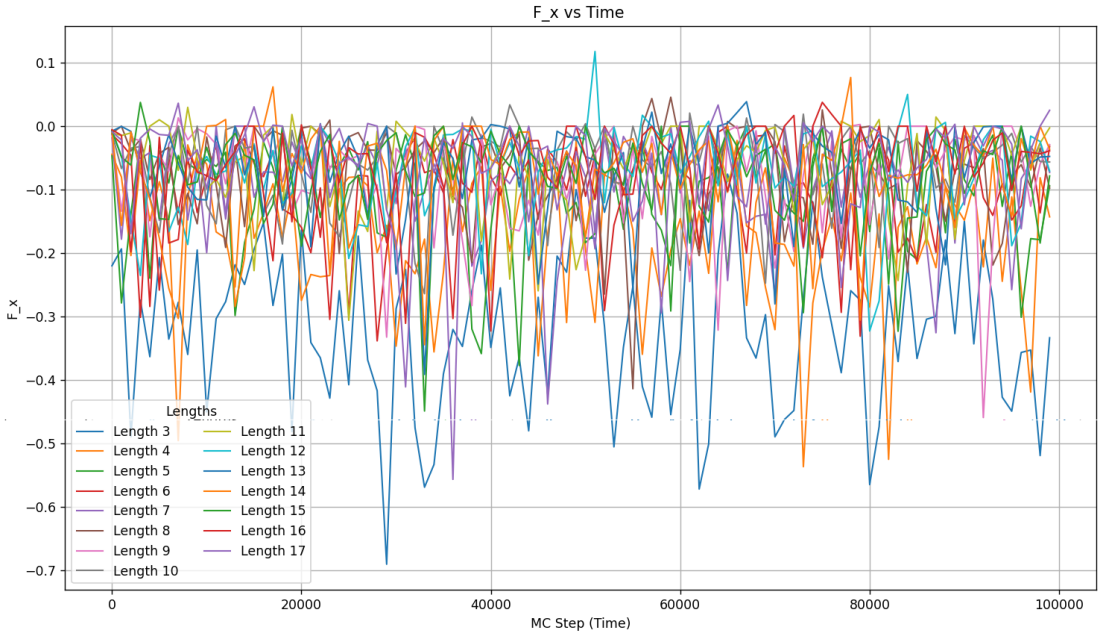


Figure 4.1: F_x vs Time graph for all lengths $3 \leq L \leq 17$.

In order to verify this mathematically, we have run a least squares regression on our interval and made sure that our regression line doesn't have a slope significantly different from zero.

Length	**Slope** ($\times 10^{-17}$)	**Error** ($\times 10^{-17}$)	**Not Significant?**
3	-0.2437	0.4608	True
4	-0.3675	0.3991	True
5	-0.1574	0.3330	True
6	-0.0001	0.2943	True
7	-0.2474	0.3520	True
8	-0.2777	0.2924	True
9	-0.3054	0.3257	True
10	0.3176	0.2461	**False**
11	0.3163	0.2704	**False**
12	0.1702	0.2629	True
13	0.2525	0.2855	True
14	0.2076	0.3211	True
15	-0.1398	0.2446	True
16	0.6619	0.2659	**False**
17	-0.0011	0.2698	True

Since lengths $L = 10, 11,$ and 16 have a regression coefficient significantly different from zero, we have to omit them from our data set. This regression was also run over several different values of time; yet, all reruns had these values significantly different from zero.

Now that we have verified proper average forces at equilibrium, we can plot our F_x average vs L plot and curve fit to find a value for the extensibility.

Based on the geometric properties of our poly-catenane structure, we expect there to be an Inverse Power Law relationship between the Average force in the x-direction and at equilibrium. Essentially as we stretch our poly-catenane structure, we expect there to be a steep increase in force for shorter lengths and a plateau for longer lengths.

We expect our relationship to follow:

$$F_x = a - bL^{-n}$$

where coefficients a, b, n are fitted to our data.

Performing an inverse power law fit for the intervals, we get the following coefficients:

$$a = -0.0680 \pm 0.0043$$
$$b = 7.3453 \pm 2.0827$$
$$n = 3.0230 \pm 0.2548$$

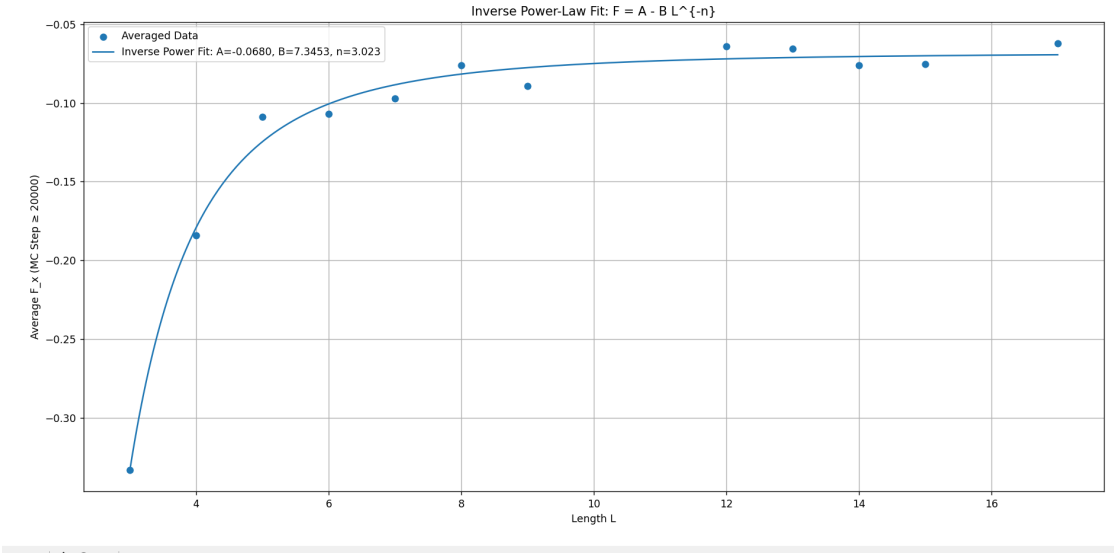


Figure 4.2: Inverse Power Law Fit for Average F_x vs Length. $R = .994$ and $R^2 = 0.9884$

Knowing that:

$$(\frac{\partial L}{\partial F_x})_N = \frac{1}{(\frac{\partial F_x}{\partial L})_N}$$

We can calculate the extensibility for these length intervals by taking the reciprocal of the partial derivative of our inverse power law.

$$(\frac{\partial F_x}{\partial L})_N = nbL^{-(n+1)}$$

So we know our extensibility follows the form:

$$(\frac{\partial L}{\partial F_x})_N = \frac{1}{nbL^{-(n+1)}} = \frac{L^{n+1}}{nb}$$

Finally, with our fitted coefficients:

$$(\frac{\partial L}{\partial F_x})_N = \frac{L^{(3.0230\pm0.2548)+1}}{(3.0230\pm0.2548)(7.3453\pm2.0827)}$$

5 Limitations of the Model and Conclusion

The model we are using has limitations in its ability to predict the thermodynamic properties in poly-catenanes. Ultimately, we would like to account for as many variables as possible and

see how they influence our extensibility. We also know that for each of our runs, the poly-catenane could be in a meta-stable state instead of equilibrium. This is most likely not the case, as the program has been run several times with different random seeds, and there is little evidence to show that the trend of the force changes significantly from run to run.

Using the Method of Metropolis Monte Carlo for the canonical ensemble of poly-catenane structures, we were able to simulate the extensibility of our poly-catenane for lengths $3 \leq L \leq 17$ to be:

$$\left(\frac{\partial L}{\partial F_x}\right)_N = \frac{L^{n+1}}{nb} \tag{5.1}$$

$$a = -0.0680 \pm 0.0043$$
$$b = 7.3453 \pm 2.0827$$
$$n = 3.0230 \pm 0.2548$$

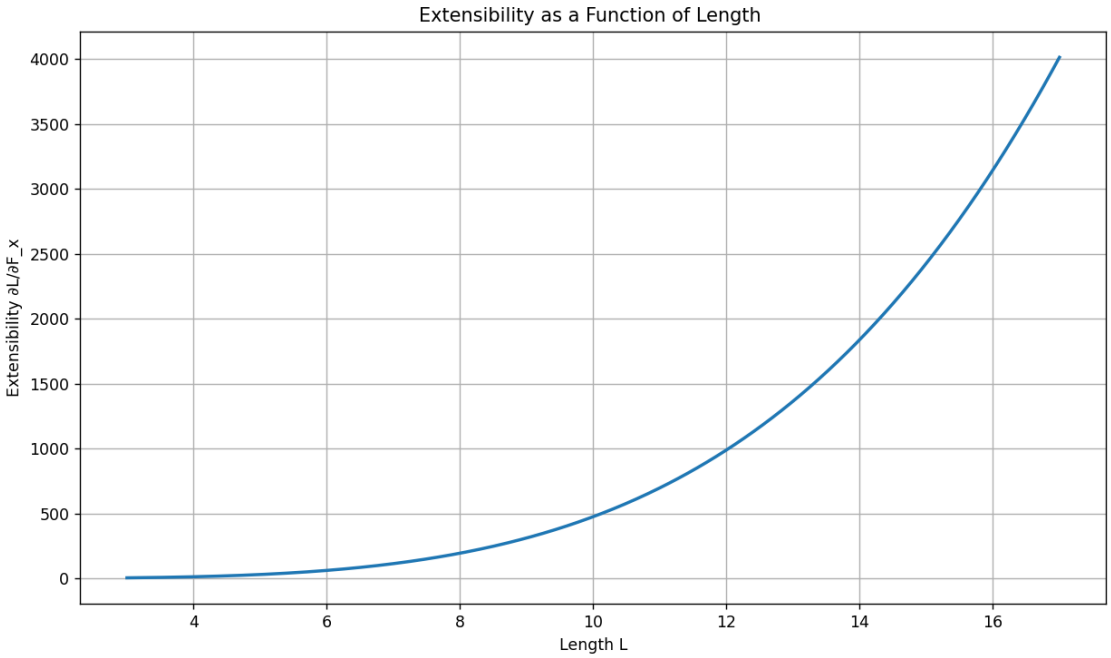


Figure 5.1: Extensibility as a Function of Length without including uncertainties.

6 References

- ReliaSoft. (n.d.). Inverse power law relationship. Retrieved December 7, 2025, from `https://help.reliasoft.com/reference/accelerated_life_testing_data_analysis/`

alt/inverse_power_law_relationship.html

- Stoddart, J. F. (2018). Lecture (pdf). Retrieved December 7, 2025, from https://www.nobelprize.org/uploads/2018/06/stoddart-lecture.pdf
- Yaoxing Chen, Dajia Xu, Yongjing Rao, and Guojie Zhang. (2024). Nonlinear Elasticity of Single Linear Polycatenane: Emergence of Stress-Softening. Retrieved from https://pubs.acs.org/doi/pdf/10.1021/acs.macromol.4c00767

Generative AI was used in the process of this research project in order to assist with rewriting and organizing C++ and Python code to display graphs and run the simulation program in a for loop. All other computations and methods were performed by me.